



上海科技大学
ShanghaiTech University

EE115 Analog Circuits 模拟电路

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Outline



- Building blocks of IC amplifier
- SEDTRA/SMITH book page 509-529;

IC design philosophy



IC design

- Resistors are replaced by transistors
- Large capacitors are avoided, and small capacitors can be fabricated.
- Low DC power supply voltage ($\sim 1\text{V}$)
- IC designer has the freedom to specify the device dimensions and to utilize device matching and arrays of devices having dimensions with specified ratios.
- Predominantly CMOS
 - » BiCMOS provides BJT



Discrete vs Integrated Circuits

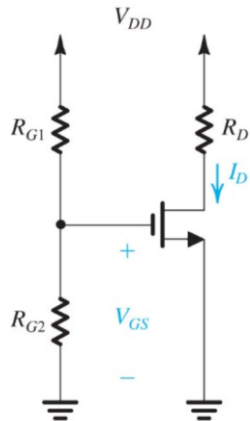
Discrete Circuits

- Resistors and capacitors are frequently used
- AC coupled with coupling capacitors
- Large DC power supply voltage
- Transistor choice limited to available parts
- Mostly BJT, some MOS

IC

- Use mostly transistors
 - » Resistors and capacitors occupy too much areas
- Mostly DC coupled (without capacitors)
- Low DC power supply voltage ($\sim 1V$)
- Can vary device size
- Predominantly CMOS
 - » BiCMOS provides BJT

Current mirror

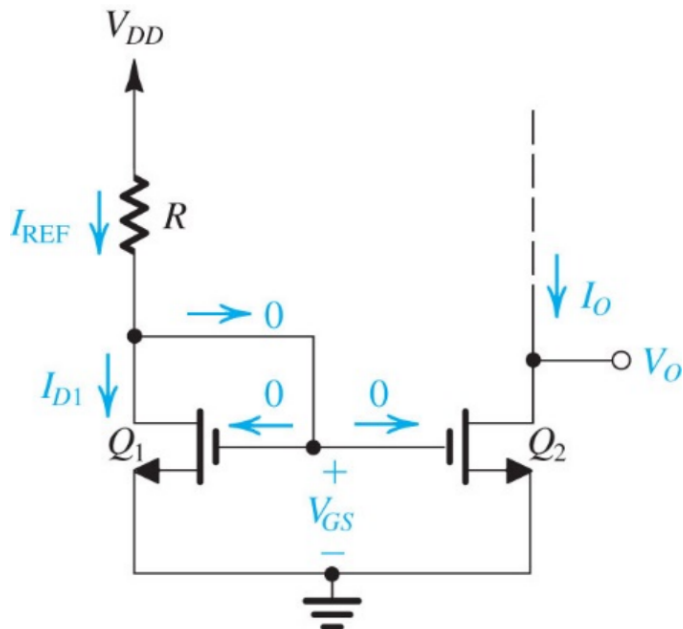


- Neglecting channel length modulation

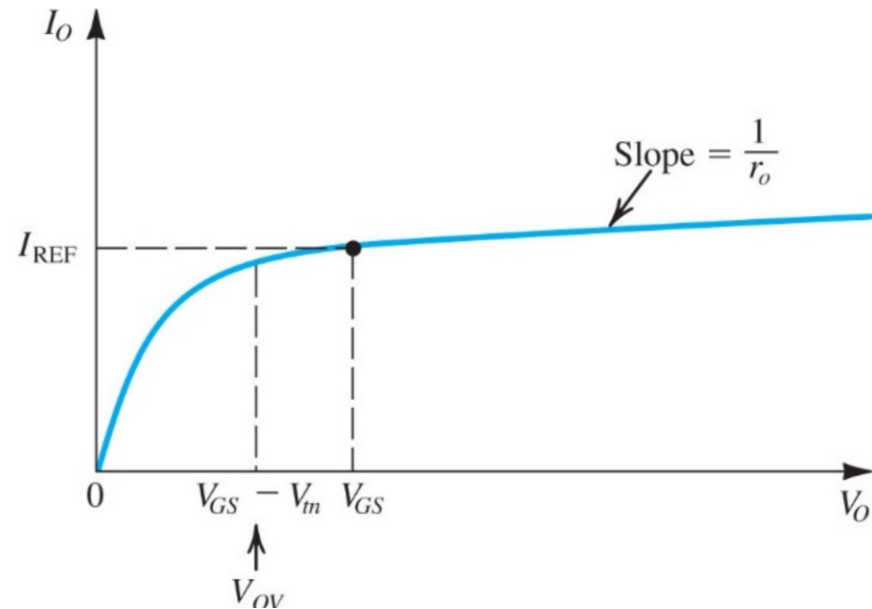
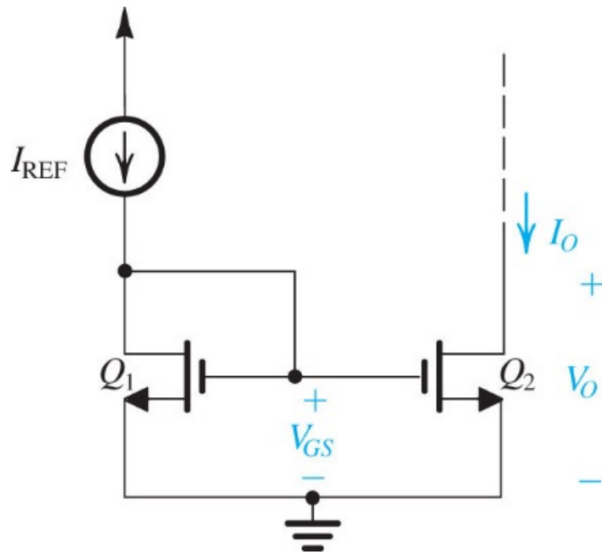
$$I_{D1} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_1 (V_{GS} - V_{tn})^2$$

$$I_O = I_{D2} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_2 (V_{GS} - V_{tn})^2$$

$$\frac{I_O}{I_{REF}} = \frac{(W/L)_2}{(W/L)_1}$$



Current Mirrors with reference source



- Neglecting channel length modulation

$$\frac{I_O}{I_{\text{REF}}} = \frac{(W/L)_2}{(W/L)_1}$$

- If considering channel length modulation

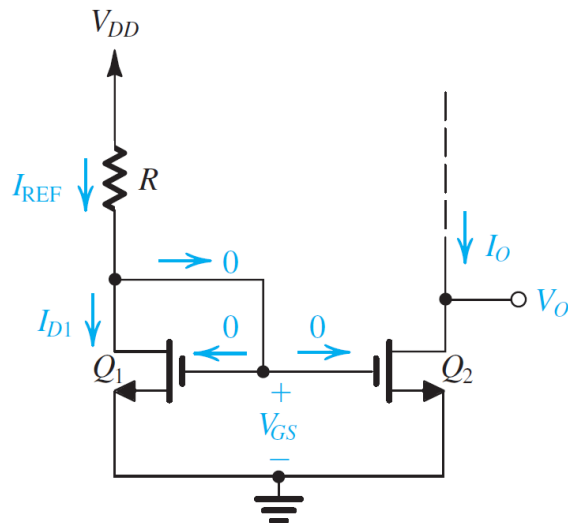
$$I_O = \frac{(W/L)_2}{(W/L)_1} I_{\text{REF}} \left(1 + \frac{V_O - V_{GS}}{V_{A2}} \right)$$

$$i_D = \frac{1}{2} k'_n \left(\frac{W}{L} \right) (v_{GS} - V_m)^2 (1 + \lambda v_{DS})$$

Example



Given $V_{DD} = 3\text{ V}$ and using $I_{REF} = 100\text{ }\mu\text{A}$, design the circuit of Fig. 8.1 to obtain an output current whose nominal value is $100\text{ }\mu\text{A}$. Find R if Q_1 and Q_2 are matched and have channel lengths of $1\text{ }\mu\text{m}$, channel widths of $10\text{ }\mu\text{m}$, $V_t = 0.7\text{ V}$, and $k'_n = 200\text{ }\mu\text{A/V}^2$. What is the lowest possible value of V_o ? Assuming that for this process technology, the Early voltage $V'_A = 20\text{ V}/\mu\text{m}$, find the output resistance of the current source. Also, find the change in output current resulting from a $+1\text{-V}$ change in V_o .



Example solution



Given $V_{DD} = 3\text{ V}$ and using $I_{REF} = 100\text{ }\mu\text{A}$, design the circuit of Fig. 8.1 to obtain an output current whose nominal value is $100\text{ }\mu\text{A}$. Find R if Q_1 and Q_2 are matched and have channel lengths of $1\text{ }\mu\text{m}$, channel widths of $10\text{ }\mu\text{m}$, $V_t = 0.7\text{ V}$, and $k'_n = 200\text{ }\mu\text{A/V}^2$. What is the lowest possible value of V_O ? Assuming that for this process technology, the Early voltage $V'_A = 20\text{ V}/\mu\text{m}$, find the output resistance of the current source. Also, find the change in output current resulting from a $+1\text{-V}$ change in V_O .

Solution

$$I_{D1} = I_{REF} = \frac{1}{2} k'_n \left(\frac{W}{L} \right)_1 V_{OV}^2$$

$$100 = \frac{1}{2} \times 200 \times 10 V_{OV}^2$$

Thus,

$$V_{OV} = 0.316\text{ V}$$

and

$$V_{GS} = V_t + V_{OV} = 0.7 + .316 \simeq 1\text{ V}$$

$$R = \frac{V_{DD} - V_{GS}}{I_{REF}} = \frac{3 - 1}{0.1\text{ mA}} = 20\text{ k}\Omega$$

$$V_{Omin} = V_{OV} \simeq 0.3\text{ V}$$

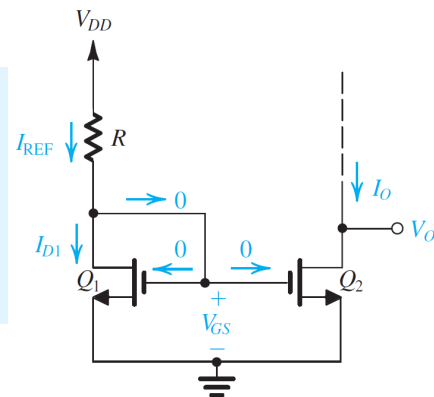
For the transistors used, $L = 1\text{ }\mu\text{m}$. Thus,

$$V_A = 20 \times 1 = 20\text{ V}$$

$$r_{o2} = \frac{20\text{ V}}{100\text{ }\mu\text{A}} = 0.2\text{ M}\Omega$$

The output current will be $100\text{ }\mu\text{A}$ at $V_O = V_{GS} = 1\text{ V}$. If V_O changes by $+1\text{ V}$, the corresponding change in I_O will be

$$\Delta I_O = \frac{\Delta V_O}{r_{o2}} = \frac{1\text{ V}}{0.2\text{ M}\Omega} = 5\text{ }\mu\text{A}$$

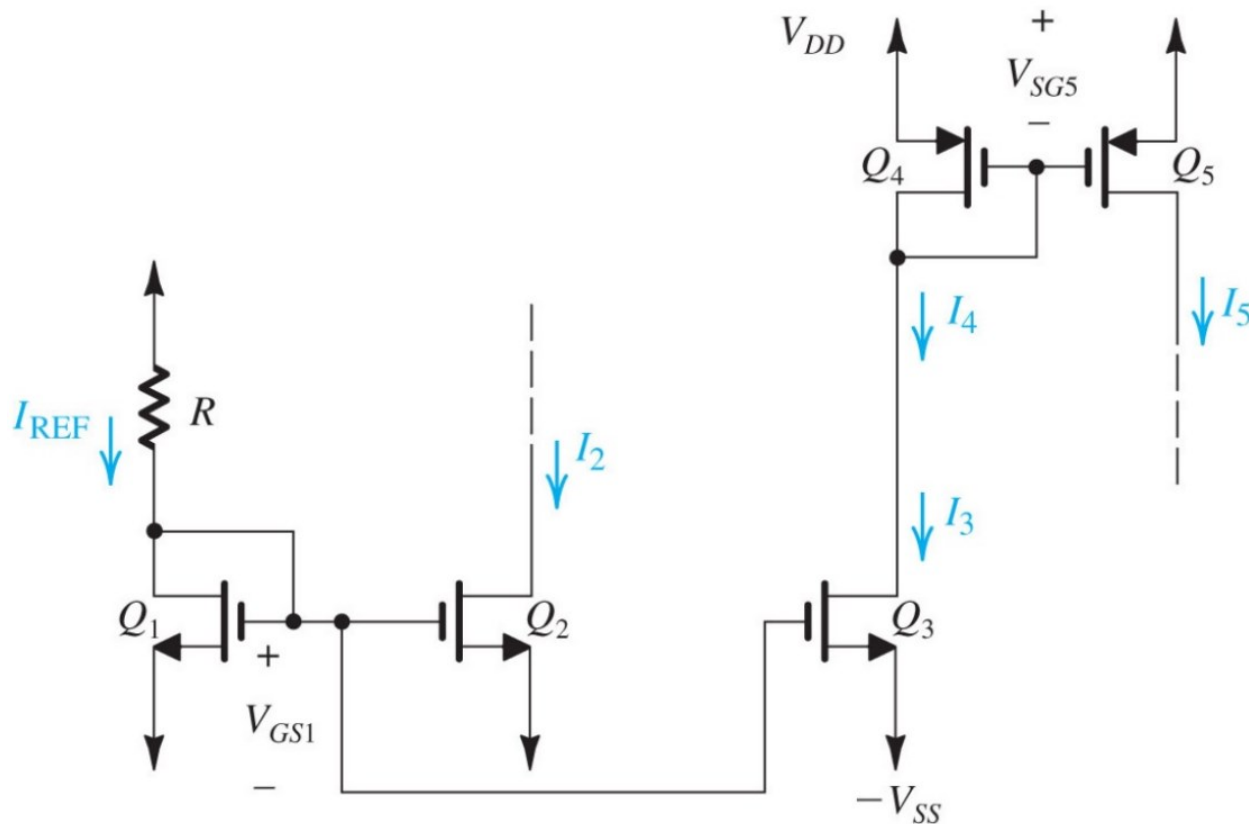


$$r_o \equiv \left[\frac{\partial i_D}{\partial v_{DS}} \right]_{v_{GS} \text{ constant}}^{-1}$$

$$= \left[\lambda \frac{k'_n}{2} \frac{W}{L} (V_{GS} - V_{tn})^2 \right]^{-1}$$

$$= \frac{1}{\lambda I_D}$$

MOS Current Steering Circuits

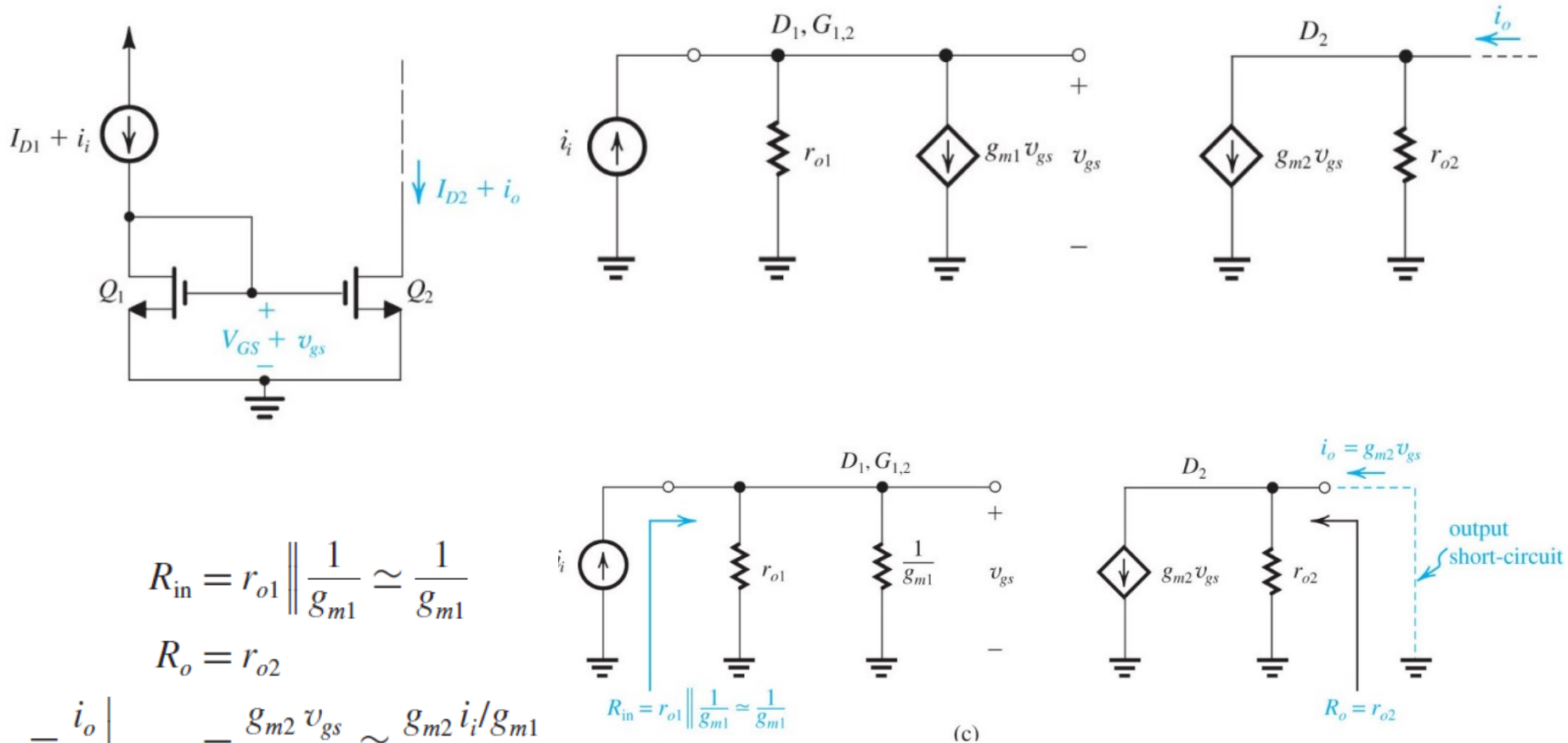


$$I_2 = I_{\text{REF}} \frac{(W/L)_2}{(W/L)_1}$$

$$I_3 = I_{\text{REF}} \frac{(W/L)_3}{(W/L)_1}$$

$$I_5 = I_4 \frac{(W/L)_5}{(W/L)_4}$$

Small-Signal Model



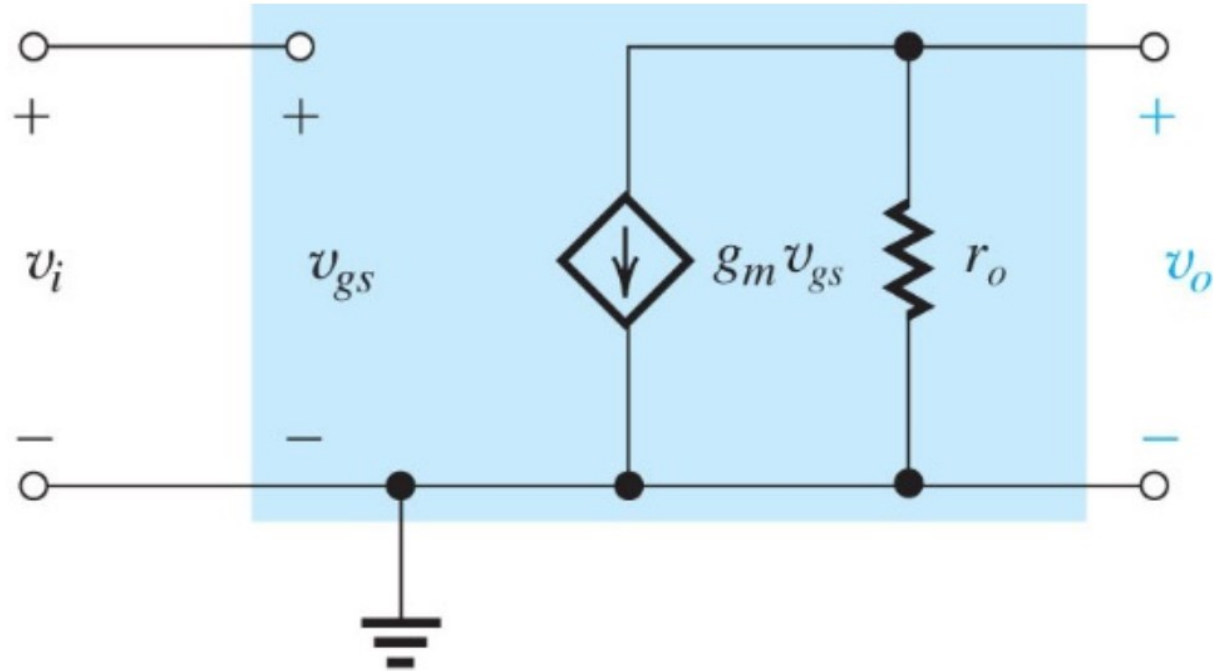
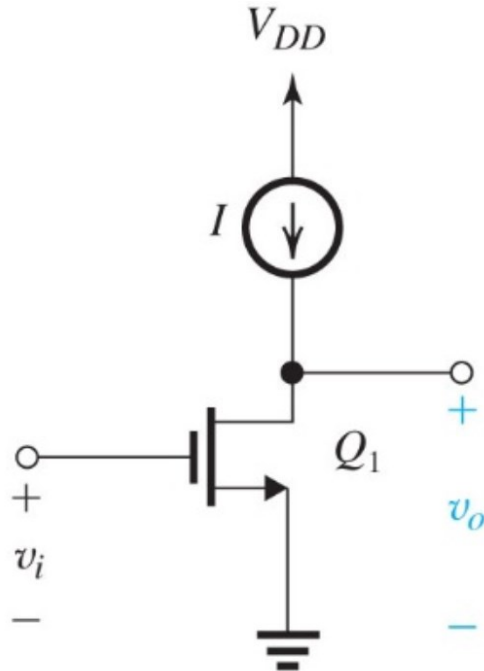
$$R_{in} = r_{o1} \parallel \frac{1}{g_{m1}} \approx \frac{1}{g_{m1}}$$

$$R_o = r_{o2}$$

$$A_{is} \equiv \frac{i_o}{i_i} \bigg|_{v_{d2}=0} = \frac{g_{m2}v_{gs}}{i_i} \approx \frac{g_{m2}i_i/g_{m1}}{i_i}$$

$$A_{is} = \frac{g_{m2}}{g_{m1}} = \frac{(W/L)_2}{(W/L)_1}$$

Basic Gain Cells with Ideal Current Source Load



$$g_m = \frac{I_D}{V_{OV}/2}$$

$$g_m = \sqrt{2\mu_n C_{ox}(W/L)}\sqrt{I_D}$$

$$r_o = \frac{V_A}{I_D} = \frac{V'_A L}{I_D}$$

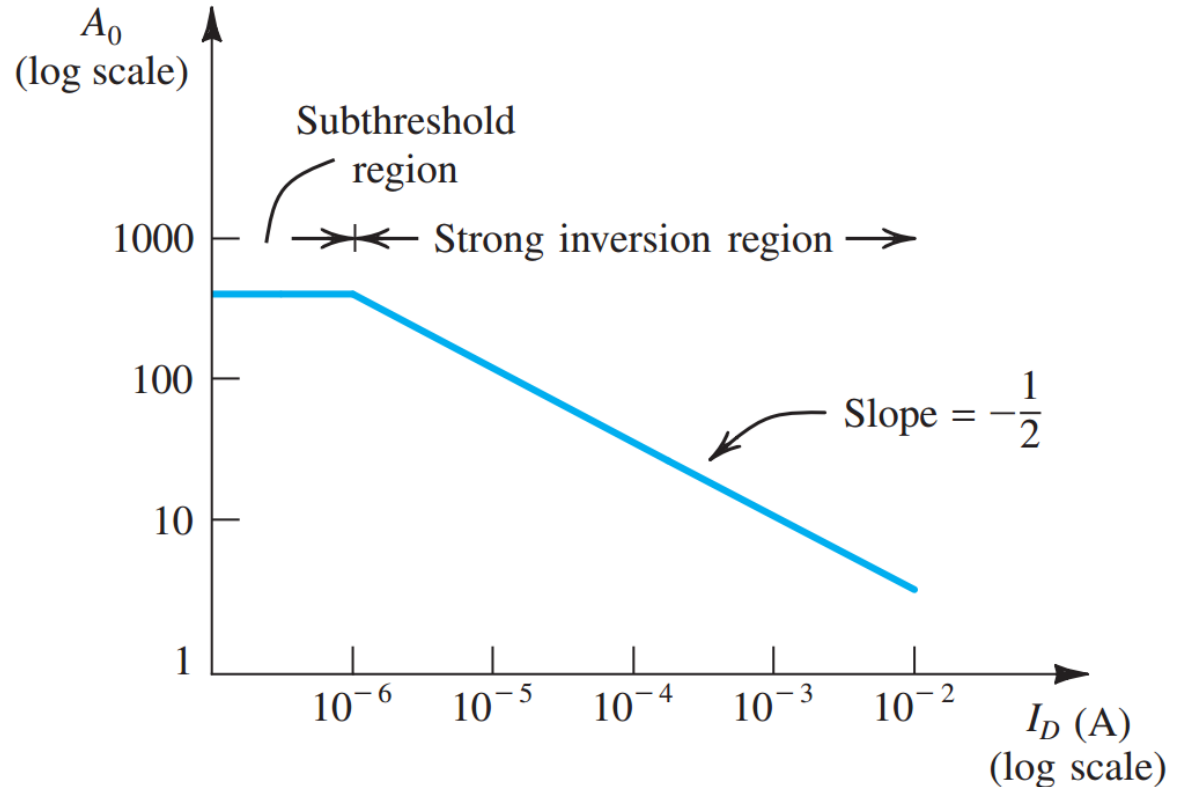
$$A_0 = \frac{V_A}{V_{OV}/2} = \frac{2V'_A L}{V_{OV}}$$

$$R_{in} = \infty$$

Intrinsic gain: $A_{vo} = -g_m r_o$

$$R_o = r_o$$

The Intrinsic Gain $g_m r_o$

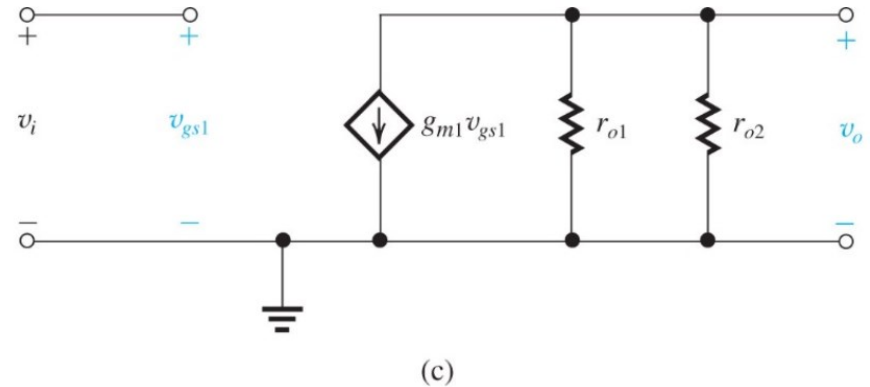
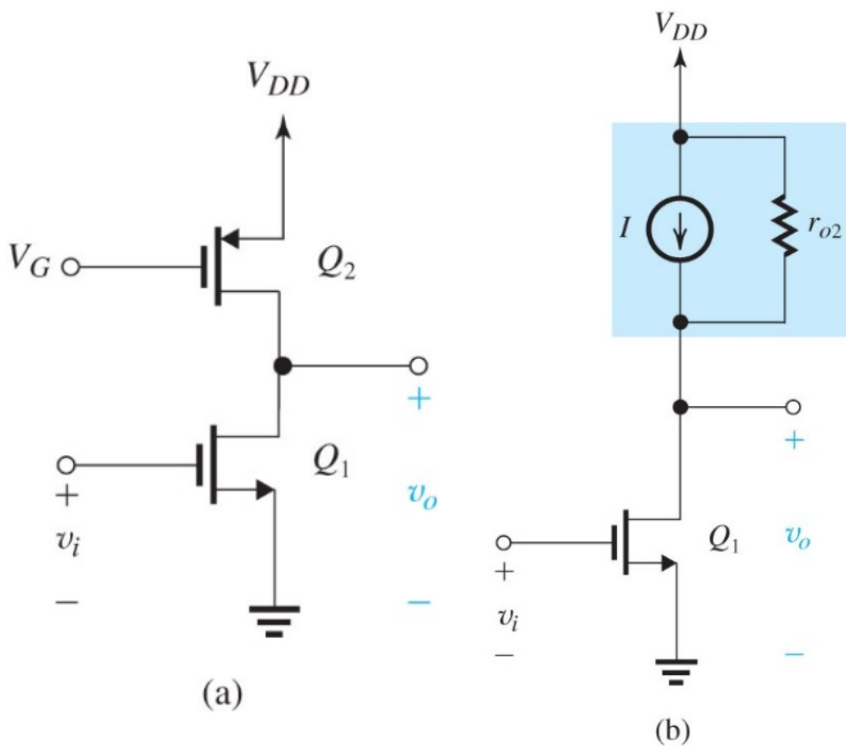


- MOSFET

$$A_0 = \frac{V'_A \sqrt{2(\mu_n C_{ox})(WL)}}{\sqrt{I_D}}$$

Smaller current gives higher gain, but paid by smaller g_m and bandwidth.

Output Resistance of Current Source Load



$$r_{o2} = \frac{|V_{A2}|}{I}$$

$$A_v \equiv \frac{v_o}{v_i} = -g_{m1}(r_{o1} \parallel r_{o2})$$

Homeworkx-due April xx 11p.m.



SEDRA/SMITH Book chapter x problem xx